

Proof of Concept Results Report

Project Name: Blockchain voting system **URL:**

<https://projectcatalyst.io/funds/11/cardano-use-cases-concept/blockchain-voting-system> **Project Number:** 1100039 **Project Leads:** David Tacuri

1. Scope of the Proof of Concept

The Proof of Concept (PoC) aimed to validate the technical feasibility of integrating blockchain technology (Cardano) into the electoral scrutiny process, ensuring traceability, integrity, and transparency.

The PoC was developed as a fully functional end-to-end system in a local environment, including:

- Digitization of records (scanning)
- Image processing and automatic recognition
- Digitization of results
- Quality control
- Blockchain registration
- Database persistence
- Evidence storage in IPFS

2. Validated System Flow

The system validated the following complete flow:

Digitization:

1. Scanning of the record
2. Recognition of the electoral board
3. Image segmentation by sections
4. Automatic recognition of votes per candidate (OCR/ML)
5. Sending images to IPFS
6. Generation of evidence hashes
7. Registration of metadata on the blockchain (Cardano)
8. Persistence in the database

Digitization and Quality Control:

1. Information request
2. Reading the image from IPFS
3. Digitization/quality control
4. Registration of metadata on the blockchain (Cardano)
5. Persistence in the database

Each transaction was successfully recorded on the blockchain, enabling complete traceability of the process.

3. Blockchain Execution

- **Network used:** Cardano Preview Testnet
- **Approximate number of transactions:** more than 1,000 transactions
- **Type of data recorded:**
 - Hashes of images (IPFS)
 - Process metadata
 - States of each phase of the record

The system allowed traceability to be consulted directly through Cardano explorers, validating the integrity of the process.

4. Performance Analysis

Significant differences were identified between the performance of traditional systems and the use of blockchain:

Type of operation	Approximate time
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Traditional transactions	< 1 second
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Transactions on Cardano	> 1 minute
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This result shows that blockchain is not suitable for real-time operations within the critical flow, but rather as a subsequent registration mechanism.

5. Metadata Management

- **Average metadata size:** < 5 KB per transaction
- **Strategy used:**
 - Use of hashes instead of complete data
 - References to IPFS for evidence storage

This maintained efficiency in the use of the blockchain.

6. Main Challenges Identified

Blockchain Latency Latency was the main challenge, with times exceeding 1 minute per transaction, limiting its use in critical real-time operations.

Signing of Multiple Transactions (CBOR) Complexity arose in managing multiple transactions generated in sequence, especially in processes requiring ordered signing and submission.

UTXO Management UTXO handling introduced additional complexity in transaction construction, perceived as a factor that slows down the process.

Evaluation of On-Chain Validations Initially, the possibility of performing on-chain validations was explored; however, due to latency and complexity, these validations were moved to the backend (off-chain), significantly improving performance.

Use of cardano-cli The use of cardano-cli proved to be an efficient and reliable tool once the strategy of avoiding complex on-chain validations was defined.

7. Critical Limitation: Scalability

One of the most important findings of the PoC is that Cardano's base layer (Layer 1) presents significant scalability limitations for this type of high-demand system, such as national electoral processes.

Considering real scenarios:

- Tens of thousands of records
- Multiple phases per record
- High volume of transactions in short time windows

Conclusion: It is not viable to operate this system directly on Layer 1 at large scale due to:

- Confirmation latency (>1 minute)
- Throughput limitations

Need for Layer 2 Solutions (Hydra) To scale this type of solution to a real environment, it is necessary to incorporate Layer 2 solutions (Layer 2), such as Hydra within the Cardano ecosystem.

Hydra would enable:

- High-speed transaction processing (near real-time)
- Significant reduction in latency

- Greater concurrent processing capacity
- More efficient costs

Under this approach, the architecture would evolve toward:

- Hydra (L2): intensive and real-time processing
- Cardano L1: settlement and final audit layer

8. Results Achieved

- Successful registration of records on the blockchain
- Complete traceability of the electoral process implemented
- Integration validated between:
 - Backend (.NET)
 - Blockchain (Cardano)
 - Decentralized storage (IPFS)
- Technical viability of the hybrid model (off-chain + on-chain) demonstrated

9. Audit Capability

The system allows:

- Verification of each stage of record processing
- Consultation of information registered on the blockchain through public explorers
- Validation of evidence integrity through hashes

This represents a significant advance in terms of electoral transparency.

10. Key Findings

- There is institutional interest in this type of solutions
- There is a significant shortage of professionals specialized in blockchain applied to government
- Institutions require working with teams or companies with proven experience, especially in critical systems

11. PoC Conclusion

The Proof of Concept demonstrates that it is technically viable to integrate Cardano blockchain into electoral processes to improve transparency and traceability.

However:

- Blockchain (L1) is not sufficient to support the real scale of the system
- It is necessary to evolve toward architectures with Layer 2 (Hydra)
- Blockchain must be used as an audit layer, not as the main transactional system

Strategic Conclusion

The PoC validates a hybrid model where:

- Traditional systems → performance
- Hydra (L2) → scalability
- Cardano (L1) → trust and audit

This approach represents a solid foundation for a real implementation in national-level electoral processes.